UnifoLM-WLA□□□□□□□□□□

□□”□□□□”□□□□□□□**□□□□□□□□□□□**□□□□□□□□□□□□□□
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- 6. □□□□□□□□□□□□□
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UnifoLM-WLA□□□□ Unitree□□□□□□□ VLA□ Vision-Language-Action□□□□□□□□□□□□□
□ Qwen3-VL□ ”□□ +□□□□□ ”□□ MMDiT□□□□□□□□□ flow-matching□□ ”□□□□□ ”□
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- □□/□□□□ 6B□□□ ~2500□□□□□□□ 64□□□□□□□□□□□□□□□□□□ REA
- □□□**□□□□□□□□□ action expert□□□□□□**□□□□□□□□□□□□□
- □□□□ starVLA□□□□□□□ Qwen-Image□ MMDiT□□□ README.md:54-57□

□□□□ □ VLM□ **□□ /□□□□□** □□□□□□□□□□□□□□□□□□□□□□□□ hidden states□□ DiT□ cross-attention□□□ key/value□□ DiT□□□□□□□□□□□□□□□ flow matching□□ 4□□□□□□□□□□□

1. □□□□

LeRobot dataloader/multi_source_dataset/ (parquet
+ mp4) 54 / 60 / / collate → dict batch
model/ QWen3.py MMDiT_ActionHeader.py
Qwen3-VL ActionEncoder + DiT + Decoder (+ robot_state) flow matching
loss last_hidden_states cross-attn K/V
: framework/VLM4A/QwenMMDiT.py
training/ train_unifolm_wla.py: VLATrainer trainer_utils/: Accelerator+DeepSp

□□□□□□□□ 36□ .py□□ ~7700□□

| □ | □□ | □□ | □□ | |—|—|—|—| | **□□** | training/train_unifolm_wla.py | 508 | main() + VLATrainer□□□□ | | **□□□□** | model/framework/base_framework.py | 269 | build_framework() □□□□□ + baseframework□□ | | | model/framework/VLM4A/QwenMMDiT.py | 386 | **□□□□□□□**□□ VLM + □□□□□□□ forward/predict | | | model/framework/share_tools.py | 559 | □□□□□ checkpoin □□□ config□□□□ | | | model/tools.py | 428 | Registry□□□□□□□□ CrossAttention□□□□ | | **VLM** | model/modules/vlm/QWen3.py | 369 | Qwen3-VL□□ + **□□□□□** + robot_state□□ hook | | | model/modules/vlm/__init__.py | 10 | □□□□□□□□□□ Qwen3-VL□ | | | model/modules/ | 239 | □□□ VLM embedding□ MLP□□□ | | **□□□** | model/modules/action_model/MMDiT_ActionHea | 367 | flow matching□□□/□□/□□/□□ | | | model/modules/action_model/DiT_modules/mmdit.py | 593 | DiT□□ + □□□□□□□ + RoPE | | | .../flow_matching_head/action_encoder.py | 84 | □□□□□□ + swish | | | .../flow_matching_head/cross_attention_dit.py | 362 | **□□□□□□□□** | | **□□** | dataloader/__init__.py | 50 | build_dataloader() □□□□ + □□□□□ | | | .../single_sour | 750 | **□□□□□**□□□□□ □ □□□□ | | | .../dataloader.py | 235 | collate_fn + DataLoader□□ | | | .../action_mapping.py | 253 | □□□□/□□□□□□□□□ | | | .../se3_utils.py | 180 | □□□□□□ + SE(3)□□□□ | | | .../stats_utils.py | 102 | □□□□□ + □□□□□□ | | | .../lerobot_wrapper.py | 235 | LeRobotDataset□□□□□ + Arrow□□ | | | .../config.py | 226 | □□ YAML□ dataclass | | | .../distributed_utils.py | 100 | □□□□□□□□□ | | | .../fps_utils.py | 83 | **□□□□□□□□□□□□** | | **□□□□** | training/trainer_utils/trainer_to | 558 | TrainerUtils□□□/□□/□□□ + □□□□□ | | | training/trainer_utils/config_tracker.py | 548 | AccessTrackedConfig + **monkey patch OmegaConf** | | | training/trainer_utils/overwatch.py | 168 | □□ + □□□ DataLoader worker□ CUDA□□ |

2. □□□

2.1 □□□□□ LeRobot□□

```
<data_base>/<data_path>/   meta/info.json # features        data/chunk-000/*.parquet
  videos/...
```

□□□□□ YAML□ unifolm_wla/dataloader/multi_source_dataset/configs/unitree.yaml□ multi_task: true□□ _load_datasets() (single_source_dataset.py:139)□ discover_task_dirs() □□ data_path □□□□ meta/info.json □□□□□□□□□□□□□ LeRobotDataset□ □□ _route_index() (single_source_dataset.py:436) □□□ index □□□□□ (□□□□ , □□ index)□

`dataloader.py:171-175`□□□□ `enabled: true`□□□□□ `ConcatDataset`□□□**□□**□□□□□ =□□□□□□□□□□□□□ `weight`□□□□□□□□□□ `dataloader.py:141-142`□□□□□□

2.2□□□□□□□□□

`_build_delta_timestamps()` (`single_source_dataset.py:121`)□

python future_ts = [i / fps for i in range(0, fps)] # fps=30 → 30   1   dt[action_key] = future_ts #   action key

LeRobot□ `delta_timestamps`□□□□□□□□□□ `(T, D)`□□”□□□ + □□ 29□”□□□ key□□ delta□ □□□□□□□□□□□

> □ `chunk_size`□□□□ 30□**□□□□□□□**□□□□□□□□ `source_fps`□

2.3□□□□□□□ 54□□ `action_mapping.py:8-43`□

□□	□□	□□	□□□□
`[0:6]`	6	□□ EE□ xyz + rotvec□	**□□□□□□□ SE(3)□□**
`[6:7]`	1	□□□	□□□□□□
`[7:13]`	6	□□□□□ 6-DoF	□□
`[13:19]`	6	□□ EE	□□ SE(3)
`[19:20]`	1	□□□	□□
`[20:26]`	6	□□□□□ 6-DoF	□□
`[26:29]`	3	waist □□	□□
`[29:32]`	3	torso □□	□□
`[32:34]`	2	□□ vx, vy	□□
`[34:35]`	1	□□ vw□□□□□	□□
`[35:41]`	6	□□□□□□□□	**□□ SE(3)**
`[41:42]`	1	□□□□	□□
`[42:48]`	6	□□□□	□□
`[48:54]`	6	□□□□	□□

□□□□ **54□ bool□□**□□□□□□□□□□□□□□□□□□□□□□□ 0□□□□ False□ □□□□□□□□ loss□ □□□□□ `MMDiT_ActionHeader.py:285-290`□□□□□□□□□

□□□□□□□□□□□ `action_mapping.py:129-138`□ ——□□□□□□□□□ §6.3□

2.4□□□□□□□ 60□□ `action_mapping.py:47-62`□

□□□□□□□□□□□□□□□□□□□□□

1. **EE□□□ rotation-6D□ 9□ = xyz + 6□□□ rotvec□ 6□□**□ `STATE_SLICES["left_xyz_rot6d"] = slice(0, 9)`□□□□ rot6d□□□□□□□□□□□□□□□□□□□□□ rotvec □□□□□□□□□□□ `matrix_to_rot6d()` (`se3_utils.py:65`)□□□□□□□□□□□□ `[R00,R10,R20,R01,R11,R21]`□
2. `[41:47]`□ `base_rotvec`□□□□□□□□□ **□□□□ xyz(3) + □□□□□ xyz(3)**□ `action_mapp`

> □ □□□ `[48:54]`□□□□**□□□□□□□□**□**□□□□□□□□**□□□□□□□□□□ 3□□□□□□□□□□□□□□□□□□□□ `docs/robot_action_state_processing.md:198` □□□□□□□□

2.5 □ □ □ □ = SE(3) □ □ □ se3_utils.py:162-180□

□ □ □ □ □ □ □ □ □ □ □ □ □ □

python T_curr = pose_to_se3_from_format(current_state_pose, ee_format) # (4,4) T_future= pose_to_se3_from_format(future_action_poses, ee_format) # (N,4,4) T_rel = se3_inverse(T_cur @ T_future # (N,4,4) return se3_to_xyz_rotvec(T_rel) # (N,6)

□ T_rel = T_curr ¹ · T_future□ **□ □ □ □ □ □ □** □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ /□ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ se3_inverse□ □ □ □ [[R , -R t],[0,1]] □ □ □ □ □ □ □ □ □ □ □ □

□ □ □ □ ee_format □ □ □ □ □ □ xyz_rpy / xyz_quat / xyz_rvec□ □ Unitree □ xyz_rpy□

2.6 □ □ □ □ stats_utils.py:38-97□

□ □ □ □ □ □ □ □ json□ □ □ dataloader/multi_source_dataset/stats/□

- **stats.json**□ □ □ □ □ □ □ □ □ □ □ □ body command□ □ min/max/mean/std/q01/q99
- **relative_stats.json**□ □ □ □ □ □ □ □ □ □ □ □ □ □ left_ee_pose_gripper_base □

□ □ □ □ □ □ □ □ □ □ □ □ □ □ normalized = (value - offset) / scale□

| □ □ | offset | scale | □ □ | |—|—|—|—| | minmax_q | (q01+q99)/2 | (q99-q01)/2 | norm_type / state_norm_type / gripper_norm_type | | zscore | mean | std | rel_norm_type□ □ □ □ □ □ □ □ | | minmax | (min+max)/2 | (max-min)/2 | □ □ |

get_normalizer □ □ □ global_*□ □ □ □ □ □ □ □ □ □ □ □ shape=(6,)□ □ □ □ □ per-step □ q01/q99□ shape=(30,6)□ □ relative_stats.json □ □ □ □ □ □ □ □ global_*□ □ □ □ □ □ □ □ §7□ □

□ □ □ □ □ □ □ □

- base_pose □ □ □ □ □ □ □ □ □ □ □ offset=0, scale=1□ single_source_dataset.py:340-342□ (TODO) HACK □ □ □
- base_rot □ □ □ □ □ 3□ □ □ □ □ □ □ □ □ □ □ □ □ □ single_source_dataset.py:344-352□

□ □ □ □ □ □ □ ”□ □ ”□ _action_norm_offset/_scale □ 54 □ □ □ □ □ □ □ □ □ □ □ □ offset/scale □ □ □ □ _build_action_norm_vectors()□ §8□ □ □ □ □ □ □ □ collate□ **□ □ □ □ □ □** □

2.7 collate□ □ □ □ □ □ □ □ □ □ □ □ □ dataloader.py:45-124□

collate_fn □ □ □ batch □ □ □ □ □ □

python { "action": (B, T, 54) # T = source_fps = 30 "action_mask": (B, 54) "state": (B, 60) "state_mask": (B, 60) "action_norm_offset": (B, 54) # ← "action_norm_scale": (B,

54) # ←   "images": {role: (B, C, H, W)} #   zeros   "image_mask":{role: (B,)} # bool "task": [str] * B "arm_type": [str] * B "robot_type":[str] * B }

□ □ □ □ □ □ □

1. **role□ □ □ □ □ □** □ all_roles□ ”□ □ □ □ □ □ ”□ □ □ □ □ □ □ dataloader.py:64-75□ □ □ □ _process_images()□ □ config.image_keys□ □ □ □ dict□ □ □ □ □ □ □ □ □ prompt □ role: <image>□ □ □ □ □ □ □ □ ——**□ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □** □
2. **□ □ role□ shape□ □ □ □** □ □ □ □ □ □ raise ValueError□ dataloader.py:98-102□ □ □ □ □ □ □ □ □ ”□ □ padding□ □ □ □ □ ”□ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □

2.8□ □ □ □ □ □ □ □ □

(a) Arrow□ □ □ lerobot_wrapper.py:145-198□ LeRobotDatasetWithSelectedKeys□ □ □ load_hf_da parquet□ □ pyarrow□ save_to_disk□ cache_dir/arrow_cache/<md5>□ □ □ .complete□ □ □ □ □ □ □ □ .tmp.<pid>.<uuid>□ os.rename□ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ load_from_disk **□ □** FileLock□ □ □ □ □ HF datasets□ CPFS□ □ stale-dentry□ □ □

(b) DataLoader worker□ □ 520 MiB□ □ □ dataloader.py:16-42 + overwatch.py:146-168□ □ □ □ □ □ □ □ □ □ □ □ □ □ ”□ □ ”□ □ □ □ □ □ □ □ □ □ □ □ □ □

1. multiprocessing_context="spawn"□ worker□ □ □ □ □ □ □ □ □ □ □
2. □ □ import□ tools.py□ □ □ □ □ overwatch = initialize_overwatch(__name__)
3. DistributedOverwatch.__init__□ □ accelerate.PartialState()□ □ torch.cuda.set_device(
4. **□ □ worker□ □ □ □ □ □ □ ~520 MiB□ CUDA context**□ □ □ □ □ □ □ □ CPU□

□ □ □ □ □ □ worker□ □ **□ □** □ sys.argv□ □ □ □ □ □ □ □ □ □ type(current_process())□ SpawnP □ □ □ re-import□ □ □ □ □ □ □ □ □ □ □ □ □ □ □ worker□ □ □ □ □ □ □ □ □ current_process().na == "SpawnProcess-N"□ overwatch.py:165-167□ □ □ □ □ □ □ □ □ □ CUDA□ PureOverwatch□ dataloader.py:33-42□ □ □ □ □ □ □ □ □ □ worker spawn□ □ □ DS_ACCELERATOR=cpu□ □ □ DeepSpeed□ gcc -laio□ □ □ □ □ □ **□ □** □ □ □ CUDA_VISIBLE_DEVICES□ □ □ □ □ □

3. □ □ □

3.1□ □ □ □ □ □ □ □

model/tools.py:117-145□ □ □ □ □ □ □ Registry□ base_framework.py:26-48□ _auto_import_frame □ **□ □ □ □** framework/□ □ □ import□ □ □ □ □ □ □ _□ □ □ □ □ □ □ □ □ @FRAMEWORK_REGISTRY. □ □ □ □ QwenMMDiT.py:176□ □

“‘python

base_framework.py:51-72

def build*framework(cfg): #□□□□ _auto*import*framework*modules() framework*id = cfg.framework.name* #□□□□□□□□ *return FRAMEWORK*REGISTRYframework_id “‘

> □□□□□*□□□□ *QwenMMDiT**□□ `config/training/unifolm_wla_pretrain_multisource.ya`
> □□□ `name: QwenGR00T` ——□□□□□□□□□□ `build_framework` □ `NotImplementedError`□
> □□□□□□□□□□□□□□□□□□□□□□□ YAML□

3.2 VLM □□□□ `QWen3.py`

`_QWen3_VL_Interface` □□ `Qwen3VLForConditionalGeneration` □□□□□□□□□□

(1)□□□□□ □ `QWen3.py:54-93`□ `flash_attention_2`□ □□□□□□□ `sdpa`□ □ `config.hidden_si` □□□ `text_config.hidden_size`□ Qwen3-VL□□ config□□ `hidden_size`□□□□ `padding_side = "right"`□

(2)□□□□□ ——□□□□□□□□□□□□□ □ `QWen3.py:216-341`□

□□□ user turn □□□□

```
 head_left: <image> cam_wrist_left: <image> cam_wrist_right: <image> Task: <instructio
State: <|robot_state_implicit_stats|><|robot_state|> Control Mode: Arms: EE, LOW BODY:
JOINT
```

- `arm_type == "dual_with_legs"` □□ , `LOW BODY: JOINT`
- □□□ `image_roles` □□□ **role □□□□□□□□□□□**
- □□□□□□□□□□□□*□□□□□□□□□□□□□ *build_vlm_dataset.py**□□□□ —□ VLM □□□□□□□□□□□□□ prompt□ □□□□□□□□□□□□

(3)□□□□□□□□□□ □ `QWen3.py:148-154`□ □□□□ `input_ids`□ □□□□ `inputs_embeds`□ □ Qwen3VL□ `get_placeholder_mask`□ **token id□ □** □□□□ token□ □□□ `input_ids`□ □□□□□ ’ embedding□□□□ ”□□□□ □□□□ special token□ embedding□ image-pad embedding □□□□□□□□□ `Image features and image tokens do not match`□ □□ `input_ids` □□ M-RoPE □□□□□□

3.3 robot_state □□□ `projector/robot_state.py` + hook

□□□□□□□□□□□□□□□□□□□**□□**□

||□ □ A□ DiT□ state token|□ □ B□ VLM□ projector||—|—|—||□ □ |`robot_state_projector.enab`
`= false`□ □ □ □ |`enabled = true`||□ □ |`state_encoder` MLP□ □ `state` `mask`(120)□ 768□ □ □ **□**
0□ token□ □ □ □ token□ |`RobotStateProjector`□ 120□ □ □ □ VLM hidden(2048)**□ □ □**
<|**`robot_state`**|> **token□ □** ||□ □ |`MMDiT_ActionHeader.py:139-147, 266-270`|`QWen3.py:95-196`
||□ □ □ |□ |`QwenMMDiT.py:199-200`□ □ □ `state_dim = 0`□ □ □ □ □ A|

□ □ B□ □ □ □ □ □ □ `QWen3.py:156-196`□ □ □ **□ □ embedding□ □ forward hook**□ □ □ □ □ □ □
`inputs_embeds`□ □ □ `input_ids`□ □ □ □ □ □ □ □ □ □ token-id□ □ □ □ □ □ □ □ □ □ M-RoPE□ □ `masked_scatter`□ □ □ □ projector□ □ □ □ □ □ □ □ hook□ □ □ □ □ □ □ □ □ □ <|`robot_sta`
□ □ □ □ □ □ □ □ □ batch size□ □ □ □ □ □ □ □ □

`RobotStateProjector`□ □ □ □ □ □ 2□ MLP□ `Linear(120→hidden) → SiLU → Linear(hidden→VLM_hi`

3.4□ □ □ □ `MMDiT_ActionHeader.py`

3.4.1□ □ □ "□ □ □ □ "

`__init__` (`MMDiT_ActionHeader.py:106-121`)□ □ □ □ □ □ □ □ □ □ □ □ □

`python config = full_config.framework.action_model # YAML   action_model_cfg = DiTConfig["D`
`#   {"input_embedding_dim":768, # "attention_head_dim":64, # "num_attention_heads":12}`
`config.update(action_model_cfg) #     action_model   config.diffusion_model_cfg.update(acti`
`#    diffusion_model_cfg diffusion_model_cfg = {**action_model_cfg, **diffusion_model_cfg}`
`#  YAML   self.model = DiT(**diffusion_model_cfg)`

□ □ □ □ □ DiT□ □ □ □ □ □ □ □ □ □ YAML□ □

|□ □ |□ |□ □ ||—|—|—||`num_attention_heads`|12|DiTConfig□ YAML□ □ □ ||`attention_head_dim`
|64|DiTConfig||`inner_dim`|**768** = 12×64|□ □ □ □ ||`cross_attention_dim`|**2048**|□ □ □ □
VLM hidden □ □ □ `QwenMMDiT.py:204-206`□ ||`num_layers`|16|YAML||`output_dim`|**1024**|
YAML|

*□ □ *input_embedding_dim=768*□ *output_dim=1024**□ □ □ □ □ □ □ □ □ □ □ □ □ □ token□ 768
□ □ □ □ □ □ □ □ □ □ □ □ □ □ 12□ ×64□ □ □ □ □ □ □ □ `action_decoder` `MLP(1024→1024→54)`□ □ □ □ □

3.4.2□ □ □ □

```
   ( ) state_encoder:  MLP(120 → 1024 → 768)    1   token    noisy_action   ActionEncoder
  +pos_embed    FiLM_pre        hidden_states (B, 1+30, 768) (B,30,108) Linear(108→768)
Embedding(1024,768)   +      (   0..29)   swish/Linear                   VLM last_hidden
                          DiT (16 × QwenImageBlock) (B, L, 2048) RMSNorm → Linear(2048→768
      + RoPE                    FiLM_dec    action_decoder MLP(1024→1024→54)    (B,30,54)
```

- `ActionEncoder`□ `MMDiT_ActionHeader.py:27-52`□ □ □ `[noisy_action ‖ mask]`(108)□ □ □ 768□ □ **□ □ □ □ □ □** □ □ □ □ □ □ □ □ □ □ □ swish + Linear□ □ □ □ □ □ □ □ `t_discretized = (t * 1000).long()`□ □ □ □ □ □ □ □ □ □ t□
- `state_encoder` □ `MLP(120→1024→768)`□ □ □ □ **60□ state□ 60□ mask = 120□ □**
  □ □ □ §4.5□ □ □

- `embodiment_embedding` + `FiLM`□ `:163-214`□ □ □ `robot_type` □ □ □ □ embedding□ □ `Linear` □ □ `(scale, shift)`□ □ `x*(1+scale)+shift` □ □ □ **□ □** □ action token □ DiT □ □ 768 □ □ □ DiT □ □ □ decoder □ □ 1024 □ □ □ □ □ FiLM □ □ **zero-init**□ AdaLN-Zero□ □ □ □ □ □ □ □ □ □ □ □ □ □ □ §6.3□

3.4.3 □ □ □ □ flow matching□ □ □ □ □ □

□ □ □ `forward`, `MMDiT_ActionHeader.py:230-291`□ □

```
python t = noise_s * (1 - Beta(1.5, 1.0).sample()) # t   (0, 0.999] noisy = (1 - t) * noise
+ t * action #     velocity = action - noise #    pred = decoder(DiT(action_encoder([noisy
‖ mask]), VLM_hidden, t)) sq_err = (pred - velocity) ** 2 loss = (sq_err * mask).sum()
/ mask.sum() #    MSE
```

`Beta(1.5,1.0)` □ □ □ □ x^0.5□ □ □ □ □ x□ 1□ □ □ `t` □ □ 0□ □ **□ □ □ □ □ □ □ □** □ `mask_part` □ □ □ `noisy` □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ ”□ ”

□ □ □ `predict_action`, `:293-350`□ □ □ □ □ □ □ □ □ □ 4 □ Euler □ □

```
python dt = 1/num_inference_timesteps # 0.25 for step in range(4): v = model(x_t, cond,
t=step/4) x_t = (x_t + dt * v) * mask #
```

`repeated_diffusion_steps = 8`□ `QwenMMDiT.py:289-305`**□ □ □ □ □ □ □ □ □ □ □ 8□ □ □ □ □ □ □ □ 8□** □ □ □ □ □ **□ (t, noise)□ □ □ □ □ □ □ □ VLM □ □ □ □ 1□ □ □ □ □ □ □ 8□ □ □ □ □ □ □ □ □ □ — —□ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □

3.5 DiT □ □ □ `DiT_modules/mmdit.py`

□ □ □ Qwen-Image □ □ □ □ □ □ double-stream□ MMDiT□

3.5.1 □ □ □ □ □ □ □ □ `QwenDoubleStreamAttnProcessor2_0`, `:169-305`□

□ □ ”□ ”□ **action**□ □ `to_q/k/v`□ □ **text**□ □ `add_q_proj/add_k_proj/add_v_proj`□ □ □ □ □ □ □ □

```
 1.      QKV   QK-norm (:218-226) 2.     RoPE (:229-234) 3.   :  joint = [action, text]
← action  ! (:248-250) 4.      flash / flash_varlen / native  (:271-285) 5.        +
(:295-303)
```

3.5.2 □ □ ”□ □ action-first”□ □ □ □ □ □ □ □ □ □ □ □ □ `mmdit.py:236-247` □ `:545-556`□

1. **FLASH_VARLEN□ key packer□** `key[b, :valid_len]` **□ □ □ □ □** □ □ □ □ ”□ □ □ □ □ □ 0□ □ □ □ □ □ □ □ □ □ □ ”□ □ □ □ □ □ □ mask □ □ assistant turn /□ padding □ □ □ □ □ □ □ □ □ □ action □ □ □ □ □ □ □ □ □ □ True □ □ □ 0-anchored □ □ □ □ □ □ □ □ □ □ □ □ □ **□ □ □ □ □ □ □ key□**
2. `QwenActionTextRotaryEmbed`□ `:46-85`□ □ □ □ □ action-first □ □ □ □ □ action □ `0..S_act-1`□ t □ `S_act` □ **action□ □ □ □ □ □ □ =action_horizon□ □ text □ □ □ □ □ /□ □ □ □ □ □** □ □ action □ □ 0□ □

3.5.3□□□□□□□□ONNX□□□□□

□□□□□□□□□□□□□□□□□□□torch dynamo / ONNX exporter□□trace□

- `pos = torch.arange(total, dtype=torch.float32)`□□□□□int64□——□ONNX exporter□□□`torch.outer`□□□□□trace□int64 Sin/Cos□□□:76-79□
- □`reshape(*x.shape[:-1], -1, 2)`□□`.flatten(3)`——`.flatten(3)`□lower□□trace□batch□□□□Reshape□:147-150□
- □□□□RoPE□`use_real=True`□□`view_as_complex`——□□□□□ONNX/TensorRT□□□□□□:62-65□
- `_attention_backend`□□□`AttentionBackendName.NATIVE`□□SDPA□□□□□□□flash-attn kernel□□ONNX□□□□:252-258□

□□□□□□□□□□□□□□/□□□□□□□□□□□□□□□□□□□□

3.5.4 AdaLN□□

`QwenImageTransformerBlock`□:307-444□□□□□□□□□`action_mod/txt_mod`□□`temb`□□`6*dim`□□□2□`3*dim`□norm1□□norm2□□□□□□□□`(shift, scale, gate)`□□□□`x*(1+scale)+shift`□`x + gate*sublayer(x)`□`_modulate`□□`[B,D]`□□□□□□□□□□`[B,L,D]`□RTC□□□□□□□□□——**RTC□real-time chunking□□□□□□□□□□□□□**□□
`temb`□`[B,D]`□

`DiT.forward`□□□AdaLN□□□`temb`□□`txt_temb`□:585□□□□□□□□□action□□

3.6□□□□ `framework/VLM4A/QwenMMDiT.py`

`forward`□□□□:233-318□

“‘python examples = *adapt*multi*source*batch(batch) # collate □ dict □ List[dict]□□ PIL □□ qwen*inputs = build*qwenvl*inputs(images, instructions, solutions=None, image*roles, arm*types) backbone*attention*mask = qwen*inputs.get(”attention*mask”) assistant*mask = qwen*inputs.pop(”assistant*mask”, None)

last*hidden = qwen*vl*interface(**qwen*inputs, robot*states=...).hidden*states[-1] # (B, L, 2048)

□ □ DiT □ key □ □ □ □ □ □ attention_mask□ assistant turn □ □ □ □

keys*attn*mask = backbone*attention*mask & ~assistant_mask

actions*target = actions[:, -action*horizon:, :] # (B, 30, 54) actions*target = cat([actions*target, mask**.**expand_as
dim=2) # (B, 30, 108)

8□ □ □

last*hidden*r = last*hidden.repeat(8, 1, 1) actions*r = actions*target.repeat(8, 1, 1) state*r = state.repeat(8, 1, 1) body*ids*r = body_ids.repeat(8)

flow*action*loss = action*model(last*hiddenr*, actions*r, stater*, keys*attn*mask*r, body*ids*r) return {”flow*action*loss”
loss, ”action_loss”: loss} “‘

`_build_state_tensor`□ :69-106□ ——□ □ □ □ □ □ □

|□ □ |□ □ ||—|—||10% | state□ state_mask□ □ □ □ ||30% |□ □ □ □ □ `N(0, 0.1)`□ □ ||60% |□ □ □ □ |

□ ”□ □ □ □/□ □ ”□ □ □ □ □ □ □ □ □ □ □ □ □ □ `training=True`□ □ □ □

`predict_action`□ □ □ □ :320-378□

□ forward □ □ □ □ - `add_generation_prompt=False`□ □ □ `<|im_start|>assistant\n`□ - □ `to_pil_preserve`□ □ `TF.to_pil_image` - □ □ □ □ `obs_image_size`□ `resize_images`□ □ □ □ □ □ □ - □ □ □ action_mask □ □ □ □ □ □ □ □ 4 □ Euler

---

4. □ □ □

4.1 □ □ □ □ □ `train_unifolm_wla.py:478-508` □ `main():441-475`□

“‘python cfg = OmegaConf.load(args.config*yaml) cfg = OmegaConf.merge(cfg, OmegaConf.from*dotlist(nor
cfg = apply*config*compat(cfg) # □ □ □ □ version_id ”0.21” schema main(cfg)

main():

cfg = wrap*config(cfg)* # [illegible] *AccessTrackedConfig accelerator = Accelerator(gradient*accumulation*steps=..*
*deepspeed*plugin=DeepSpeedPlugin()) output*dir = setup*directories(cfg) vla = build*framework(cfg)* #
[illegible] *vla*train*dataloader = prepare*data(...) optimizer, lr*scheduler =*
*setup*optimizer*and*scheduler(vla, cfg) trainer = VLATrainer(...); trainer.prepare_training(); trainer.train()
“‘

`apply_config_compat()`[illegible] `share_tools.py:418-524`[illegible]
YAML [illegible]

|[illegible] |[illegible] ||—|—||`action_horizon`[illegible] `future_action_window_size`|[illegible]
`future = horizon - 1`[illegible] warning[illegible] `action_horizon`[illegible] ||`diffusion_model_cfg.output_`
|[illegible] `action_model.hidden_size`[illegible] ||`diffusion_model_cfg.cross_attention_dim`|[illegible]
`framework.qwenvl.vl_hidden_dim`[illegible] ||`action_model.action_hidden_dim`|[illegible] `hidden_size`
[illegible] ||`past_action_window_size`|[illegible] 0[illegible] 0[illegible] YAML[illegible]
||`version_id`|[illegible] `"0.21"`|

4.2 `AccessTrackedConfig`[illegible] ”[illegible]”[illegible] `config_tracker.py`[illegible]

[illegible]
[illegible] `__getattr__`/`__getitem__`/`get()`[illegible] `export_accessed_config()`[illegible]
`config.yaml`[illegible] checkpoint[illegible] ”[illegible]”[illegible] `from_pretrained`[illegible]

[illegible]

1. `_children`[illegible] config[illegible] wrapper[illegible] `__setattr__`[illegible] stale wrapper[illegible]
2. `merge_framework_config`[illegible] `share_tools.py:230-248`[illegible] `framework`[illegible] *[illegible] *_local_accessed**[illegible] `build_framework`[illegible] `name`[illegible] `config.yaml`[illegible]
3. [illegible] `config_tracker.py:486-549`[illegible] **monkey patch**[illegible] **5**[illegible] **OmegaConf**[illegible] [illegible] `to_container`/`save`/`to_yaml`/`is_config`/`merge`[illegible] `AccessTrackedConfig`[illegible] —[illegible] import[illegible] omegaconf[illegible]

4.3 `VLATrainer.prepare_training()`[illegible] `:120-159`[illegible]

[illegible]

1. `set_seed(cfg.seed + rank)` ——[illegible]

2. `_save_initial_configs()` ——□□□ **config.full.yaml**□ **config.yaml**□ □□□□□□□□□□ checkpoint□□ / DeepSpeed□□□ / dataloader□□□□□ □□□ run□□□□□□□□□□ `from_pretrained`□
3. `_init_checkpointing()` ——`is_resume`□□□□ checkpoint□□□□□ `pretrained_checkpoint`
4. `_adjust_lr_scheduler_for_resume()` ——□ `for _ in range(completed_steps): lr_scheduler.` □□
5. `freeze_backbones()`□ `freeze_modules`□□□□□□□□□□□□□
6. `print_trainable_parameters()`
7. `accelerator.prepare(model, optimizer[, dataloader])`
8. `_init_wandb()`□ `mode="offline"`□

4.4□□□□□□ :308-352□

```
python while completed_steps < max_train_steps: batch = self._get_next_batch() # StopItera
    iterator + set_epoch step_metrics = self._train_step(batch) if accelerator.sync_gradien
completed_steps += 1 #       if completed_steps % eval_interval == 0: step_metrics = eval_a
_log_metrics(step_metrics) if completed_steps % save_interval == 0 and completed_steps
> 0: _save_checkpoint()
```

`_train_step`□ `:385-416`□ □

```
python with accelerator.accumulate(model): optimizer.zero_grad() with torch.autocast("cuda"
dtype=torch.bfloat16): output_dict = model.forward(batch) total_loss = output_dict["action_
accelerator.backward(total_loss) if gradient_clipping: accelerator.clip_grad_norm_(...)
optimizer.step() if accelerator.sync_gradients: # ←    lr_scheduler.step()
```

`sync_gradients`□□□□□□□□□□□□□□□□ scheduler□□ `gradient_accumulation_steps` □□□□□□□ warmup□□□□□ cosine□ `max_train_steps`□□□□□□ `min_lr`□

4.5□□□□□□□□□□ `trainer_tools.py:88-144`□

`build_param_lr_groups`□ `cfg.trainer.learning_rate`□□ dict□

```
yaml learning_rate: base: 1.0e-04 #   qwen_vl_interface: 1.0e-05 # VLM   1/10    action_mod
1.0e-04 #
```

□□□□ `base`□ key□□ `getattr`□□□□□□□□□□□□□□□□ param group□□□□□□□ `base`□□ `freeze_modules`□□□□□□□□□□□□□□ VLM□□□□□□□□□□□□□□□□

4.6□□□ checkpoint

- `initialize_overwatch()`□ `overwatch.py:146`□□□ `DistributedOverwatch`□□□□ INFO□ □□□□ ERROR□□ `PureOverwatch`□□□□ / spawn worker□
- checkpoint□□ `steps_{N}_pytorch_model.pt`□ `steps_{N}_model.safetensors`□ `_get_latest_c` □□□□□□□□□□□□ `trainer_tools.py:520-522`□

- □□□□□□□□□ `final_model/`
- □□□□□ `summary.jsonl`

4.7 □□ YAML □ DeepSpeed □□

`config/training/unifolm_wla_pretrain_multisource.yaml` □□□□ `action_horizon: 30`□ `action_d`
54□ `state_dim: 60`□ `repeated_diffusion_steps: 8`□ `per_device_batch_size: 8`□ `gradient_accumu`
4□ `max_train_steps: 200000`□ `num_warmup_steps: 2000`□ `save_format: safetensors`□ `freeze_modu`
`''`□□□□□□□

`config/deepseeds/` □□ 6 □□□□□□□□□□

- `ds_config.yaml` / `ds_config_debug_cpu_offload.yaml` / `zero3.yaml` —— □ **DeepSpeed json**
- `deepspeed_zero2.yaml` / `deepspeed_zero3.yaml` / `zero2.yaml` —— **Accelerate □□□□□** □ `distributed_type: DEEPSPEED` + □□□□□ json□

□□□ `ds_config.yaml`□ **ZeRO stage 2**□□□ 3□□ `bf16: true`□ `allgather/reduce_bucket_size: 5e8`□ `overlap_comm` + `contiguous_gradients`□ stage 2 □□ 3□□□□□□□ —— VLM 4B □□□ 8□□ ZeRO-2 □□□□□□ stage-3 □□□ all-gather □□□

---

5. □□□ Trace□□□□□

□□□□□□□□ `action_horizon=30`□ DiT-B□ Qwen3-VL-4B□ batch=1□□□□

| # | □□ | □□ | □□ ||—|—|—|—|| 1 | `LeRobotDataset.__getitem__` | action `(30, D_raw)`, state `(D_raw,)` | D_raw □□□□□□ || 2 | `compute_relative_actions` | `(30, 6)` | □ /□□□□□ || 3 | `map_action_chunk` | `(30, 54)` + `(54,)` bool | □□□□ || 4 | `map_state` | `(60,)` + `(60,)` bool | rot6d □□ || 5 | `collate_fn` | action `(B,30,54)`, state `(B,60)`, images `{role:(B,3,336,448)}` ||| 6 | `_adapt_multi_source_batch` | `List[B]` of `{image:[PIL], lang, action, state, ...}` | □□□□□ || 7 | `build_qwenvl_inputs` | `input_ids (B,L)`, `pixel_values`, `attention_mask` | L ≈ 3×~100(□) + ~40(□□) || 8 | `qwen_vl_interface(...)` | `hidden_states[-1] = (B, L, 2048)` | □□ hidden □□ logits || 9 | `_build_state_tensor` | `(B, 1, 120)` | 60 state □ 60 mask || 10 | `actions_target` + mask □□ | `(B, 30, 108)` | 54 □□□ 54 mask || 11 | ×8 repeat | `(8B, 30, 108)`, `(8B, L, 2048)` ||| 12 | `ActionEncoder` | `(8B, 30, 768)` | 108 □ 768 || 13 | `state_encoder` | `(8B, 1, 768)` | 120 □ 1024 □ 768 || 14 | `hidden_states` | `(8B, 31, 768)` | 1 state token + 30 action tokens || 15 | `DiT` □□□□ | text `(8B, L, 2048)` □ RMSNorm □ Linear □ `(8B, L, 768)` ||| 16 | 16× □□□□□□□□ | action □ `(8B,31,768)`□ text □ `(8B,L,768)` | joint □□ 31+L || 17 | `proj_out_2` | `(8B, 31, 1024)` | 768 □ 1024 || 18 | `action_decoder` | `(8B, 31, 54)` | 1024 □ 1024 □ 54 || 19 | `pred_actions = pred[:, -30:]` | `(8B, 30, 54)` | **□□□ 30 □ token** || 20 | □□□□ MSE vs `velocity` | scalar ||

□ 19 □□□□□ □□□□ 14 □□ state token □□ **□□□** □□□ ”□□ 30 □ ”□□□ action □□□□□ `pred[:, -H:]` □□ `pred[:, :H]` □□□□

---

6. □□□□□□□□□□□□□

6.1 □□□ VLM □□□□□□□□□□□□□□

`QwenMMDiT.forward` □□ `build_qwenvl_inputs(..., solutions=None)`□ □□ `QWen3.py:316` □ `if solutions is not None` □□□□□□ *□□□ `labels`□□□ *LM-CE loss*□□□ `assistant_mask`*□ VLM □□□□□□□ DiT □ cross-attention□

□□□□□□□□□□□□□/□□□□□□□□ teacher-forcing□□□□□□ □□ LM head □ tokenizer □□□□□□□□□□□□□□□□□□□□□ flow matching □□□□□□

□□□□□ ”□ VLM□□□□□□□ /□ CoT”□□□□□□□□□□ `forward_vlm()`□ `compute_loss(`
□□□□□□□□ `base_framework.py:137-203`□□□□□□□□□□ **□□ trainer□□□□□□**□□
`model.forward(batch)`□ □

6.2 □□□□ MMDiT □□□□□□□□ cross-attention□

□□ DiT□ cross-attention□□□□□□□□□□□□ K/V□□□□□□□□ MMDiT□□□ **□□/□**
MLP□□□ □□□□□□□□□□□□□ ”□□ ”□□□ `DiT.forward`□□□□□□□ text□□□□
action □□□□ text □□□□□□□□□□□□□□ □□ ”□□ token □□□□ 31□□□□ token □□ L□ ”□□□□□□

6.3 □□□□□ embodiment embedding□ □ `MMDiT_ActionHeader.py:82-97` □□□

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> Single-arm sources (bridge/fractal/libero/...) all map into the SHARED left action slot, so without an embodiment signal the shared decoder averages their conflicting distributions and **the left arm collapses to ~0**.

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□□□□ DiT□□ `robot_type → embedding`□□□□□□□□□□ encoder/decoder□□□□□□□
□ DiT □□□□ `DEFAULT_EMBODIMENT_TYPES` □□□□ **□□□□□** □□□ append-only□ □ index 0 □□□ ”unknown”□ `build_embodiment_map` □□ `i+1`□ □

6.4 □□□ DiT □ cross*attention*dim □□□□□□□□

`QwenMMDiT.py:204-206` □ `self.qwen_vl_interface.model.config.hidden_size` □□ YAML □□
□□ VLM □□□□□□□ Qwen3-VL-2B/4B/8B hidden □□□ 2048/2048/4096 □□□□□□□□□□

6.5□ □ □ □ `spawn`□ □ □ `fork`□

`dataloader.py:203-205`□ □ □ fork□ □ □ worker□ □ □ trainer□ □ □ □ □ □ CUDA context□ □ □ □ □ □ CPU-only□ □ □ CUDA context□ □ □ □ □ □ fork+CUDA□ □ □ □ □ □ □ □ □ □

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7. □ □ □ □ □ □

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7.1□ `config/training/*.yaml`□ `framework.name`□ `QwenGR00T`□ □ □ □ □ □ `QwenMMDiT`

`unifolm_wla_pretrain_multisource.yaml:9`□ `QwenGR00T`□ `build_framework`□ □ `NotImplementedErro` `Framework 'QwenGR00T' is not implemented. Available frameworks: ['QwenMMDiT']`□ **□ □ □ □ □ □**

□ □ `configs/unitree.yaml:121`□ WBT□ □ □ `data_path`□ □ □ □ `UnifoLM_G1_Dex1_Dataset/`□ □ `docs/train_action_expert.md:41,50`□ □ □ □ WBT□ `UnifoLM_WBT_Dataset`□ □ □ □ □ □ □ □ □ □

7.2□ □ □ /□ □ □ DiT key□ □ □ □ □ □ □ □ □ □ □ generation prompt□

□ □ □ □ □

1. `QwenMMDiT.forward`□ `solutions=None`□ `add_generation_prompt = (solutions is None) = True`□ `QWen3.py:300-301`□
2. □ □ **□ □** □ prompt□ □ □ `...Control Mode: ...<|im_end|>\n<|im_start|>assistant\n`
3. `assistant_mask`□ `None`□ `:316`□ `if solutions is not None`□ □ □ □ □ □ □ `QwenMMDiT.py:272` □ `~assistant_mask`□ □ □ □ **□ □ □ □**
4. `predict_action`□ `add_generation_prompt=False`□ `QwenMMDiT.py:341`□ □ **□ □** □ key□ □ □ □ `...<|im_end|>`

□ □ □ □ □ □ DiT□ □ □ □ □ 3□ token□ `QWen3.py:293-299`□ □ □ □ □ □ □ □ □ ”□ □ □ □ solutions□ assistant turn□ mask□ □ □ □ □ □ □ □ □ header□ □ □ □ ”——□ □ □ □ □ □ □ QwenMMDiT□ □ □ □ □ □ □ □ □ □

□ □ □ □ □ □ □ 3□ key□ □ □ □ □ □ □ **□ □ /□ □ □ □ □ □** □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ `add_generation_prompt=False`□ □ □ `forward`□ □ □ □ solutions□ □ □ □ □ □ □

7.3□ `action_horizon`□ □ □ chunk□ □ □ □ □ □ □

`actions_target = actions[:, -self.action_horizon:, :]`□ `QwenMMDiT.py:277`□ **□ □ □** □ □ chunk □ □ □ □ `source_fps`□ =30□ □ `chunk_size`□ □ □ □ □ □ □ □ □ □ □ □ □ □ □ YAML□ `action_horizon=3`

□□□□ source_fps=30□□□□□□□□□□□□□□ action_horizon□□ 8□□□□□□□ t=22..29 □ 8 □□□□□□□□ 1/4□□ □□□□□□□□ ”□ 8 □ ”□ **□□□□□** chunk_len >= action_horizon □□□□□□

7.4□ `target_fps` / `source_fps` □□□□□□□□□

fps_utils.py□□ resample_action_chunk□ build_bspline_resample_matrix**□□□□□□□** □ _bui □□ source_fps□□□□□□ target_fps□□□□□□ log□□ single_source_dataset.py:118□ □ □□□□□□□ 10 FPS □□□□□□□□□ chunk□ 10 □□□□□□ 1 □□□ 30 FPS □□ 30 □□ 1 □ —— □□□ action_horizon=30 □□□□□□□□□□□ □ 10 □□□□ LeRobot □ delta □□ pad/□□□ 30 □□□□□□□□□□□□□□□□□

7.5□ `eval_action_model` □□□□□□□ batch

train()□□ eval_interval□□□ eval_action_model()□□□□□ examples = self._get_next_bat —**□□□□□□□□□□□□ batch□□□** □□□□□□□□□□□□ eval□□□□□□□□ ”□□□□ +□□□□□□□□□□ □□□□□□□ 500□ □□□□□□□□□□□□□□□□□□□□□ eval dataloader□

7.6□ □□□□□□□□□□□□□

main()□□□ train_unifolm_wla.py:474-475□ □□□□□ dist.barrier()□ dist.destroy_process □□□□□□□□ dist.barrier()□□ RuntimeError: Default process group has not been initialized□ □□ _log_metrics□□ dist.get_rank()□ :281□ □□□□□□□□□ **□□□ torchrun / accelerate launch □□** □□□ python train_unifolm_wla.py□

7.7□ `unitree.yaml` □ `precollected_stats_path` □□□ CWD □□□□□□

precollected_stats_path: "unifolm_wla/dataloader/multi_source_dataset/stats"□ unitree.yaml □ Path(...) □□□□□ **□□□□□□** □□□ data_base□ YAML□ □□□□ docs/train_action_ex □□□□□□□□□□□□□□□□□□□ ——□□□□□□□□□ FileNotFoundError□ □□□□□□ YAML □□□□ data_base □□□

7.8□ numpy 2.x + `fastparquet` □□□□□□□

pyproject.toml □□□□ numpy==2.2.6□ pyarrow==21.0.0□ fastparquet==2024.11.0□ torch==2.8.0 transformers 5.x□ □□□□□□□ QWen3.py:68 □□□ dtype=torch.bfloat16□ 5.x□ API□ □□□ torch_dtype□ □ □□□□□□□□ 5.x □□□ **□□□□ transformers**□

7.9□ collate□ PIL□□ tensor□□□□□

□□□□□ `_process_images`□□ tensor□ `collate_fn` stack□ `(B,C,H,W)`□ `_adapt_multi_source_ba`
□ `TF.to_pil_image`□□ PIL□ processor□□□ tensor□ `_build_state_tensor`□□ state□ list□□ tensor□

□ `(B,3,336,448)`□□□□□□□□□□□ GPU□ CPU(PIL)□ CPU tensor□□□□□□□ role □□□□□□□□□□□□□ tensor□□□□□□□□□ PIL□□□□ processor□**□□□□□□□**

7.10□ `_load_stats`□□□□□□ `precollected_stats_path`

`single_source_dataset.py:268-273`□□ `raise ValueError`□□□□□□□□□ `peer_data_paths` □ `merge_left_right_ee_stats`□□□□□ `config.py`□□□□□□ `unitree.yaml`□□□□□□**□□** in the removed multi*source*dataset.py"□□□□□□□□□□□□□□□□□□□□□□□□ □□ `docs/robot_action_state_processing.md` §12/§13□□□□□□□□□□□□□**□□□□□□**

7.11□ □□□□□

- `baseframework.__init__(self, hf_config=PretrainedConfig())`□□□□□□□□ `base_fra` config□□
- `share_tools.py:99` `isinstance(x, OmegaConf.__class__)`□□□□□□□□□□□□□ "typical not hit"
- `trainer_tools.py:137` `ReferenceError(...)`□□□□□□□□□□□□□□□ `raise`□
- `train_unifolm_wla.py:39`□ `load_fast_tokenizer()`□□□□□□□□□ FAST tokenizer □□□□
- `train_unifolm_wla.py:91`□ rank□□□□ `dist.is_initialized() and dist.get_rank()==0`□ □ `:281`□□□ `dist.get_rank()`□□□□□□

8.□□□ /□□□□□

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|□□ /□□ |□□ |□□ ||—|—|—||model/modules/action_model/flow_matching_head/cross_attenti |362|**□□□□□□□**□□□ QwenGR00T□□ LayerwiseFlowmatchingActionHead□□□ || dataloader/multi_source_dataset/fps_utils.py|83|**□□□□□□□**□□ §7.4□ ||model/tools.py: FrameworkTools|~90|□□□□ base_framework.py□ import□□□□□□□□□ unnormalize_actio □ gripper_channel_idx=6□□□□□□ 54□□□□□□□□□□□□□□ ||model/tools.py: CrossAttention|~62|□□□□□□ VGGT□□□□□□ forward□ docstring□□□ vggt_feature□ ||model/tools.py: preprocess_images|~90|□□□□ Prismatic/OpenVLA□ 518px□□□□□□ ||model/tools.py: auto_get_module_keys|22|□□□ ||model/tools.py: print_freeze_status |33|□□□ ||share_tools.py: read_model_config|57|□□□□□ read_mode_config□□□□□□ YAML□□ JSON□ ||share_tools.py: populate_layerwise_dit_cfg|35|□□□□□ LayerwiseFlow □□□ || share_tools.py: state2str_transform / add_discretized_state_to_instruction | ~30|□□□□ π_0.5□□□ "□□□□□□ prompt"□□□□□□ ||share_tools.py: NamespaceWith |~40|□□□□□□□□□□□□ ||projector/robot_state.py: add_robot_state_projector

/ patch_robot_state_forward / _prepare_robot_state_inputs / get_llm_model / get_llm_input_embeddi
| ~120 | □□□□□□□ ms-swift □□□ ”patch model.forward”□□□ QWen3.py □□□□□□
forward-hook□□□□□□□ RobotStateProjector□ load_robot_state_projector_state_dict
||trainer_utils/trainer_tools.py: setup_optimizer_and_scheduler|33|□□□□ train_unifolm_w
□□□□□□□ fused=True□□□□□□□ ||trainer_utils/trainer_tools.py: eval_qwenpi
/ compute_grad_angle_with_stats / pcgrad_project / only_main_process / is_main_process /
extract_json_from_string|~120|□□□□ PCGrad□□□□□ IoU□□□□□□□□□□□□
||base_framework.py: compute_loss / supports_training_tag / forward_vlm | ~66 |□□□ ”□
tag□□ loss□□ ”□□□□ vla / vlm / world□□□□□□□ trainer□□□ model.forward()□□□□□
||training/train_unifolm_wla.py: load_fast_tokenizer | 2 |□□□ || action_mapping.py:
map_single_left_action|17|□□□ ||config.py: ImageKeyConfig.resize / DatasetSourceConfig.im
| — |□□□ deprecated ||DatasetSourceConfig.peer_data_paths / merge_left_right_ee_stats
| — |□□□□□□□□□□□ || collate □□ action_norm_offset / action_norm_scale | — |
□□□□□ collate□ **□□□□** ||state_unnorm | — |□□□□ collate_fn□□□□□□□□ ||
action_model.action_hidden_dim | — |□□□□ apply_config_compat□□□□□□□□□□□
|

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9. □□□□□

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1. README.md + docs/train_action_expert.md —— □□□□□□
2. docs/robot_action_state_processing.md §1–§3 —— **□□/□□□□□□□□□** □□□□ ”54/60 □ ”□□□
3. action_mapping.py —— □□□□□□□□□□□□ slice
4. se3_utils.py:162-180 —— □□□□□ 10□
5. single_source_dataset.py:451-530 —— __getitem__□ 8□□□
6. dataloader.py:45-124 —— collate□□□ batch□□
7. MMDiT_ActionHeader.py:230-350 —— □□ forward + □□□□□ **□□□□□□□□**
8. QwenMMDiT.py:233-378 —— □□□□□□□□□□
9. train_unifolm_wla.py:308-416 —— □□□□

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- **□□□□□** □ docs/robot_action_state_processing.md §4–§16 + stats_utils.py + lerobot_wrap
- **□□□□□** □ mmdit.py□ QwenDoubleStreamAttnProcessor2_0 + QwenImageTransformerBlock□ □□ Qwen-Image □□□□□□□□
- **□□□□** □ overwatch.py + dataloader.py:_CpuAcceleratorWorkerDataLoader + config_tracker.
- **□□□□□□□** □ embodiment_embedding + FiLM + DEFAULT_EMBODIMENT_TYPES

□ □ □ □ □ □ □ □

“‘python

1) □□□□□□□□□□□□□

from unifolm*wla.dataloader.multi*source*dataset.dataloader import create*training*dataloader dl, ds* = *create*training*dataloader(”.../unitree.yaml”, batch*size=2, num_workers=0) b = next(iter(dl)) for k, v in b.items(): print(k, v.shape if hasattr(v, ”shape”) else (type(v), len(v) if hasattr(v, ”**len**”) else v))

□□□□：action (2,30,54), action_mask (2,54), state (2,60), images {role:(2,3,336,448)}

2) □ □ □ □ □ □ □ □ □ VLM

import torch from types import SimpleNamespace

□ □ □ fake full_config□ □ □ MMDiTFlowmatchingActionHead.forward

□ □ : vl_embs (B,L,2048), actions (B,30,108), state (B,1,120) □ loss □ □ □

3) □ getitem □ □ □ □ □ □ □ §5 □ □ □

“‘

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10. □ □ □

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1. □ □ □ □ □ □ rotation-6D □ □ □ □ □ □ rotvec□ □ □ □ □ □ □ □ □ □ □ □
2. `[48:54]` □
3. `T_rel = T_curr`^{1} `· T_future` □
4. 54 □ □ □ □ □ □ □ □ □ □ □ □ loss □ □ □ □ □ □ □ □
5. `repeated_diffusion_steps=8` □ □ □ □ □ □ □ □ □ □ □ □
6. □ □ □ `t` □ □ □ □ □ □ □ □ □ □ □ □ `noise_s` □ □ □ □ □ □
7. □ □ □ □ `t=0` □ □ `t=1` □ □ □ □ □ □ □ □ □ □
8. □ □ □ DiT □ □ □ □ □ □ □ `[action, text]` □ □ □ `[text, action]`□ □ □ □ □ □ □
9. RoPE □ □ □ □ action □ □ □ □ 0□
10. `embodiment_embedding` □ □ □ □ □ □ □ □ □ □ □ □ FiLM □ zero-init□
11. □ □ □ □ □ Qwen3-VL □ □ □ □ □ `input_ids` □ □ □ □ □ `inputs_embeds`□
12. robot_state □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □
13. `lamp/sync_gradients` □ □ □ □ □ □ □ □ □ □
14. `AccessTrackedConfig` □ □ □ □ monkey patch OmegaConf □ 5 □ □ □ □
15. DataLoader worker □ □ 520 MiB□ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ `CUDA_VISIBLE_DEVICES` □ □ □
16. `read_mode_config` □ □ checkpoint □ □ □ □ □ □ □ □ □
17. □ □ □ □ □ □ YAML □ □ □ □ □ □ □ □ □
18. `pred[:, -H:]` □ □ □ □ □ □ □ □ □ □ □ □

□ □ □ □ □ □

54 (xyz+rotvec EE/ SE(3)) 60 (EE xyz+rot6d [41:47] +) chunk
= source_fps 30 chunk_size minmax_q(/ /) + zscore( ) global_* VLM Qwen3-VL-
last_hidden_states cross-attention VLM hidden → RMSNorm → Linear(2048→768) + state
token DiT DiT-B: 12 × 64 = inner 768 16 output 1024 token ActionEncoder(108→768)
+ pos_embed(1024,768) + FiLM_pre MSE(pred_velocity, action - noise) t ~ 0.999*(1-Beta(
4 Euler dt=0.25 Accelerate + DeepSpeed ZeRO-2 + bf16 action_horizon=30 repeat=8
LR base 1e-4 / VLM 1e-5 / action_model 1e-4 cosine_with_min_lr